

Exploring the role of sex hormones and gender diversity in multiple sclerosis

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Abstract

Sex and sex hormones are thought to influence multiple sclerosis (MS) through effects on inflammation, myelination and neurodegeneration, and exogenous hormones have been explored for their therapeutic potential. However, our understanding of how sex hormones influence MS disease processes and outcomes remains incomplete. Furthermore, our current knowledge is derived primarily from studies that focus exclusively on cisgender populations with exclusion of gender-diverse people. Gender-affirming hormone therapy comprising exogenous sex hormones or sex hormone blocking agents are commonly used by transgender and gender-diverse individuals, and it could influence MS risk and outcomes at various stages of disease. A better understanding of the impact and potential therapeutic effects of both endogenous and exogenous sex hormones in MS is needed to improve care and outcomes for cisgender individuals and, moreover, for gender-diverse populations wherein an evidence base does not exist. In this Perspective, we discuss the effects of endogenous and exogenous sex hormones in MS, including their potential therapeutic benefits, and examine both established sex-based dimorphisms and the potential for gender-diverse dimorphisms. We advocate for future research that includes gender-diverse people to enhance our knowledge of the interplay of sex and sex hormones in MS, leading to the development of more effective and inclusive treatment strategies and improvement of care for all individuals with MS.

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Introduction

Multiple sclerosis (MS) exhibits notable sex-based dimorphisms, characterized by a higher prevalence among people assigned female at birth and by a higher frequency of progressive disease in people assigned male at birth¹. These variations among individuals are influenced by a combination of sex hormones, genetics, and environmental and lifestyle exposures². Although our understanding of the influence of exogenous sex hormones on MS outcomes is expanding, knowledge remains incomplete, limited by small cohorts or underpowered trials in cisgender populations. The therapeutic potential of exogenous hormones in cisgender populations at different reproductive life stages is actively being explored and needs ongoing, in-depth investigation to optimize care for all individuals with MS. To complement this research, and to enhance inclusive clinical care, we must also examine the impacts of exogenous hormones on MS outcomes in gender-diverse individuals.

Transgender and gender-diverse individuals (hereafter referred to as trans) are individuals whose gender differs from the sex they were assigned at birth and include people with binary (male or female) and non-binary identities (Table 1). Current studies suggest that trans representation among people with MS is 0.45%, which is consistent with rates in the general population³. The MS International Federation estimates that 2.8 million people are living with MS⁴; therefore, approximately 12,600 individuals with MS could identify as trans.

Gender-affirming therapy (GAT) aims to align the physical characteristics of an individual with their gender identity and can encompass surgical and hormone (gender-affirming hormone therapy (GAHT)) interventions. GAT alleviates gender dysphoria, which is defined as psychological distress owing to gender incongruence, and improves psychological functioning^{5,6}. The number of people seeking GAT has surged over the past decade, both locally in Australia and globally^{7–11}. However, trans populations have been excluded or not identified in the vast majority of health-care research to date, which includes only binary outcomes and assumes cisgender identity, creating a knowledge gap in understanding how GAHT and gender diversity impact MS outcomes¹².

Research on the effect of gender and GAT on MS is challenging owing to the small trans population in this rare and heterogeneous disease, alongside the variability in individual hormone profiles and GAHT regimens. In this Perspective, we outline the current, limited knowledge base on this topic. We explore the role of sex hormones in modulating inflammation, myelination and neurodegeneration in MS, with a focus on their interactions with the immune system and CNS. We further delve into the effect of exogenous sex hormones in MS, including their potential benefits, drawing on evidence derived from both preclinical and clinical studies¹³. Despite the above challenges, we call to action the research community to consciously include trans individuals by asking about gender identity and sex assigned at birth in their research. This inclusion will deepen our understanding of how chromosomes, sex hormones and MS pathology interact, leading to more effective and inclusive treatments for all.

A brief overview of endogenous sex hormones

The ovaries predominantly produce oestrogen – including oestradiol, oestriol (elevated during gestation) and oestrone – and progesterone, alongside a small amount of testosterone¹⁴. The testes serve as the primary producers of androgens, whereas oestrogen (primarily oestradiol) can be generated through the aromatization of testosterone. Aromatization specifically refers to the enzymatic conversion of testosterone into oestrogen¹⁴. Furthermore, circulating testosterone is converted via 5- α -reductase to dihydrotestosterone (DHT), which

exhibits heightened androgenicity and exclusively acts on androgen receptors, resisting aromatization¹⁴.

Sex hormones exert their effects through specific sex hormone receptors, including oestrogen receptors (ER) ER α and ER β , progesterone receptors¹³ and androgen receptors. These effects can be mediated through classical pathways involving direct receptor activation wherein ligand–receptor interactions trigger rapid cellular responses¹⁵. Alternatively, non-classical pathways, such as transcriptional modulation, gene methylation and other indirect mechanisms, can lead to more gradual and sustained effects. Both pathways have the capacity to operate independently or in combination¹⁵.

The distribution of sex hormone receptors determines their specific effects; receptor expression is upregulated or downregulated in response to changes in hormone concentrations¹⁶. This intricate interplay, coupled with factors such as chromosomes, genetics, age, environment and lifestyle exposures, probably contributes to sex-specific effects in MS.

The link between sex hormones and MS
A lens on the immune system

MS is a complex condition characterized by both autoimmune inflammatory activity and neurodegeneration¹⁷. Inflammatory processes cause CNS lesion formation that can manifest clinically as relapses. Both adaptive and innate immune cells – predominantly T and B lymphocytes – orchestrate a complex inflammatory cascade resulting in demyelination and axonal damage¹⁸. Sex hormone receptors expressed on immune cells exhibit a nuanced duality, manifesting both modulatory and inflammatory effects¹⁴ (Fig. 1 and Box 1).

Sex dimorphisms are observed with respect to immune mediated inflammatory activity. Epidemiological studies indicate a gender bias in MS prevalence, with higher rates in individuals assigned female at birth that has increased over time, with geographical variations potentially owing to environmental factors¹⁹. This disparity emerges after

Table 1 | Definitions of gender identities

Gender identity	Definition
Cisgender	Individuals whose gender is congruent with their sex assigned at birth (i.e. not transgender)
Transgender or trans	Individuals whose gender is incongruent with their sex assigned at birth
Transgender or trans men	Individuals assigned female at birth who identify as men
Transgender or trans women	Individuals assigned male at birth who identify as women
Gender fluid	Individuals whose gender fluctuates and is incongruent with their sex assigned at birth (see non-binary)
Non-binary	An umbrella term for any number of gender identities that sit within, outside of, across or between the spectrum of the male and female binary; a non-binary person might identify as gender-fluid, gender-queer, trans masculine, trans feminine, agender, bigender or other
Agender	Someone who does not identify with any gender
LGBT+	Lesbian, gay, bisexual, transgender, queer or questioning, intersex, asexual, and other sexual and gender minority identities

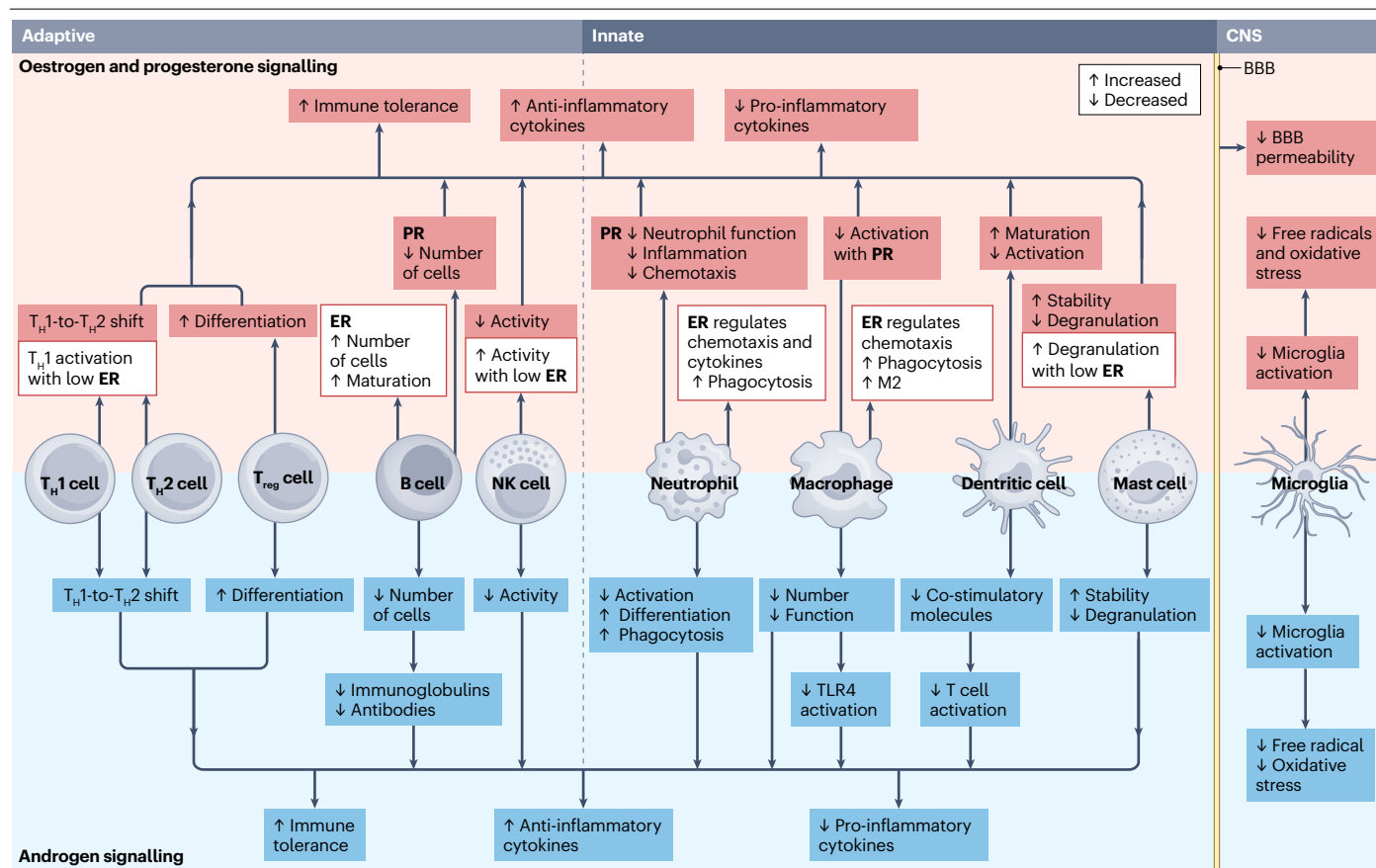


Fig. 1 | Sex hormone receptors and the immune system. The figure summarizes the effects of increased signalling of androgen, oestrogen and progesterone receptors on the innate, adaptive and CNS immune system responses. Evidence suggests that enhanced signalling from the hormones generally modulates immune responses, resulting in reduced inflammatory processes that are

associated with multiple sclerosis (MS)^{14,181,182}. Text in white boxes indicates potential inflammatory effects of oestrogen receptor signalling. BBB, blood–brain barrier; ER, oestrogen receptor; M2, M2 anti-inflammatory macrophage; NK cell, natural killer cell; PR, progesterone receptor; T_H1 cell, T helper 1 cell; T_H2 cell, T helper 2 cell; T_{reg} cell, regulatory T cell; TLR4, Toll-like receptor 4.

menarche and is not observed in children, with puberty considered a risk factor for individuals assigned female at birth^{19–23}.

During reproductive years, people assigned female at birth, particularly those with relapse-onset MS, exhibit a more inflammatory phenotype, with higher relapse rates and gadolinium-enhancing lesions on MRI compared with similar aged individuals assigned male at birth^{24–26}. These observations are supported by pathological findings that show heightened T cell reactivity to proteolipid protein and more pronounced cytokine secretion in response to myelin proteins compared with healthy controls^{27–29}. Despite the evidence of increased inflammation, people assigned female at birth reach disability milestones at a later chronological age than individuals assigned male at birth^{30–34}. Sex differences in MS disease trajectory are less prominent in late-onset MS (age >50 years) in the perimenopausal or postmenopausal period³⁵.

Oestrogen is thought to have a biphasic dose effect, being immunostimulatory at low levels (menstrual cycling concentrations) and anti-inflammatory at higher levels (gestational concentrations)^{36–40}. Hormonal shifts during pregnancy create an immunotolerant environment; oestrogens are present at higher concentrations during pregnancy than any other life period⁴¹. Progesterone, testosterone and prolactin also increase, but to a lesser extent, and peak in the third

trimester⁴². These shifts suppress MS-associated inflammatory activity during pregnancy, but reduction in the concentrations at delivery leads to increased inflammatory activity postpartum^{43,44}. Our understanding of the influence of pregnancy on longer-term MS outcomes is evolving, with potential effects also shaped by placental factors and processes beyond sex hormones^{45,46}.

A lens on the central nervous system

Sex hormones permeate the CNS wherein they act on hormone receptors that are widely distributed throughout. These receptors are notably concentrated in the frontal cortex, limbic system and cerebellum and are present on both neurons and glial cells^{47,48}. Evidence from both clinical studies and experimental animal models underscores the considerable effect of sex hormones on cognition and psychology, and their role in neuroprotection (Fig. 2 and Box 1).

In MS, neurodegeneration is probably caused in part by compartmentalized intrathecal, peri-lesional and non-lesional parenchymal immune responses, together with immune-independent processes. These processes lead to axonal dysfunction, mitochondrial failure and cell death, all resulting in axonal and neuronal loss and increased disability⁴⁹. Evidence suggests that the progression and outcomes of the neurodegenerative

processes in MS display sex dimorphisms that might be driven by differences in hormone levels in the CNS. People assigned male at birth have higher tendency to present with motor and cerebellar symptoms, more severe MRI features, including brain atrophy and T1 black holes, and

more smouldering (chronically active lesions) and grey matter lesions on pathology than individuals assigned female at birth^{50–53}. Moreover, people assigned male at birth also have more severe neurocognitive deficits^{54,55} and increased frequency of progressive disease^{24,54–59}.

Box 1 | Summary of the effect of sex hormones on the immune system and CNS

Oestrogen

Immune effects

- Cytokine regulation: mostly reduces T helper 1 (T_H1) cell responses and associated pro-inflammatory cytokine expression (such as tumour necrosis factor (TNF), IL-1, IL-6, IL-12 and interferon- γ (IFN γ)); stimulates T helper 2 (T_H2) cells and anti-inflammatory cytokine production (such as IL-10, IL-4 and TGF β)¹⁸¹
- Immune modulation: promotes anti-inflammatory T cells (T_H2 cells and regulatory T (T_{reg}) cells)¹⁸¹
- Potential inflammatory actions: promotes B cell differentiation; can protect autoreactive B cells from negative selection; lower levels of oestrogen can have stimulatory effects on T_H1 cell responses¹⁸¹

CNS effects

- Anti-inflammatory action on astrocytes and microglia¹⁸³
- Neural cell metabolic regulation and increases glutamate uptake¹⁸⁵
- Oestrogen receptor (ER)- β ligand promotes remyelination¹⁸⁶
- Proliferation of neural progenitor cells in dose dependent pattern¹⁸³
- Blood–brain barrier regulation¹⁸⁷

Progesterone

Immune effects

- Cytokine regulation: reduced pathways to production of pro-inflammatory cytokines¹⁸⁸

- Immune modulation: promotes an anti-inflammatory response, affecting T cell differentiation and inhibiting natural killer cell production of IFN γ ¹⁸⁸

CNS effects

- Increases CNS cell survival, activating anti-apoptotic mechanisms¹⁸⁹
- Bioenergetic system regulation¹⁸⁹
- Promotes oligodendrocyte progenitor proliferation and myelin synthesis (more consistently than oestrogen)¹⁸⁹
- Regulatory effect on glial cells¹⁸⁹

Androgens

Immune effects

- Cytokine regulation: modulate innate immune cells to reduce pro-inflammatory cytokine production^{122,190–192}
- Immune modulation: promote T cell shifts (T_H1 to T_H2) and self-tolerance^{191,193–195}

CNS effects

- Neuroprotective and trophic effects on neurons, inducing differentiation and promoting neurite growth^{122,196}
- Safeguards against glutamate-induced toxicity and oxidative stress^{197,198}
- Glucose and metabolic pathway homeostasis^{199,200}
- Prevention of demyelination; stimulates oligodendrocytes and remyelination^{116,201}

System	Cell type	Oestrogen	Progesterone	Androgens
Immune system	T _H 1 cells	↑ if oestrogen ↓; ↓ if oestrogen ↑ ¹⁸²	↓ ²⁰²	↓ ^{191,193}
	T _H 2 cells	↑ ¹⁸¹	↑ ²⁰²	↑ ^{191,194}
	T _{reg} cells	↑ ¹⁸¹	↑ ²⁰²	↑ ¹⁹⁵
	B cells	↑; ↑ B _{reg} cells ^{181,203}	↑; ↑ B _{reg} cells ²⁰³	↓ ↑ B _{reg} cells; ↑ if low androgens ^{204,205}
	Natural killer cells	↓ ^{181,206}	↓ ²⁰⁷	↓ ²⁰⁸
	Neutrophils	Regulates chemotaxis and infiltration ¹⁸¹	–	↑ Differentiation ²⁰⁹
	Macrophages	Regulates chemotaxis and phagocytic activity; ↑ M2 (refs. 181,210)	↓ Activation; ↑ M2 (ref. 210)	↓ TLR4 (ref. 211); ↑ M2 (ref. 209)
	Mast cells	Mixed ²¹²	Stabilize ²¹³	Modulate ²¹⁴
CNS	Dendritic cells	↑ Maturation; regulation of cytokines and chemokines ^{215,216}	↑ Maturation; ↓ activation; ↓ inflammatory cytokines ^{217,218}	Modulation; ↓ inflammatory cytokines ¹⁹²
	Microglia	↓ Activation; modulation ¹⁸³	↓ Activation; modulation ^{219,220}	↓ Activation ^{221,222}
	Oligodendrocytes	↑ Survival; ↑ production of myelin ¹⁸⁶	↑ Survival; ↑ myelin production; ↑ myelin repair ^{105,223}	Stimulatory ^{116,201}
	Astrocytes	Modulation ^{221,224,225} ; glutamate regulation ¹⁸⁵	Modulation; ↓ inflammation ¹⁰⁵	Modulation ^{221,226}

This is a simplified table, and many of the changes mentioned are context dependent. For more detailed reviews on the effects of sex hormones on the immune system and within the CNS, see refs. 14,59,181–184,227–229. B_{reg} cell, regulatory B cell; M2, M2 anti-inflammatory macrophage; TLR4, Toll-like receptor 4.

Active neurodegenerative processes in MS are hypothesized to be linked to androgen decline associated with ageing, though the exact mechanisms involving sex hormones remain unclear. Additionally, immunosenescence and inflammaging – immune processes that have a substantial role in MS progression – are observed more frequently in individuals assigned male at birth compared with those assigned female at birth^{60,61}. However, direct evidence of sex dimorphism in immunosenescence within the MS population remains to be established. Furthermore, whether one sex is related to a more severe MS

phenotype than the other is debatable. No difference is observed in the sex ratios among individuals with primary progressive MS, whereas a higher prevalence of people assigned female at birth is observed in relapse-onset MS⁶². Additionally, no sex differences are observed in MS outcomes in those with primary progressive MS⁶².

Gender dimorphism in MS

Sex dimorphisms in MS between cisgender individuals are influenced by sex hormone interactions with receptors in the CNS and immune cells,

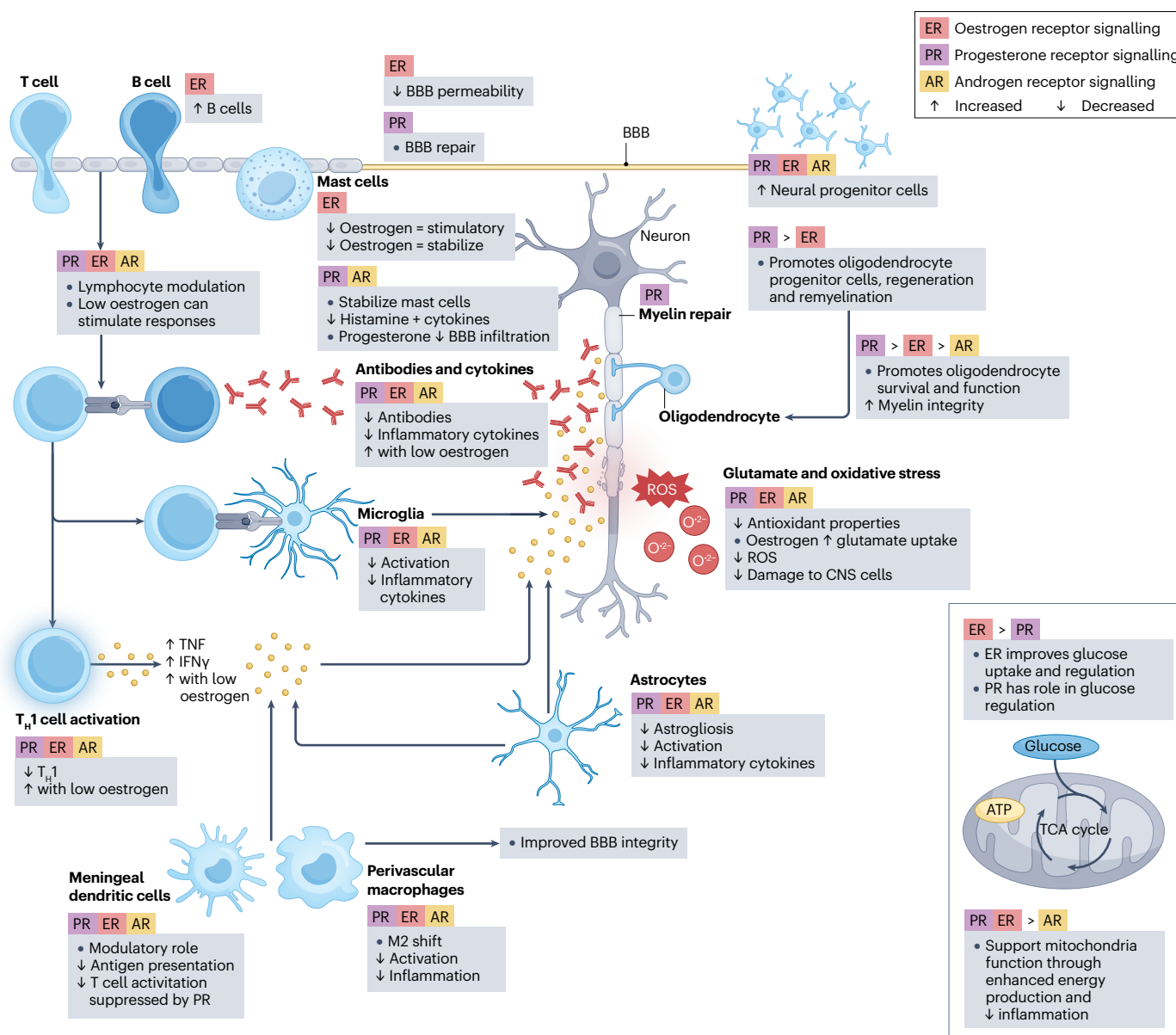


Fig. 2 | The neuroprotective role of sex hormone receptors in MS. The figure illustrates the multifaceted roles of sex hormones oestrogen, progesterone and testosterone in modulating neuroinflammation, neuroprotection and neuroregeneration in multiple sclerosis (MS)^{59,183,184}. The effects of these hormones, mediated by their respective receptors, are highly context-dependent and vary according to cell type, immune environment and the physiological state of the individual. As many of these effects are based on preliminary research,

further studies are required to elucidate their precise impact and mechanisms. The relative dominance and comparative strength of receptor signalling in neuroprotection is indicated by 'PR > ER > AR', wherein progesterone receptor (PR) signalling has the greatest impact. AR, androgen receptor; BBB, blood–brain barrier; ER, oestrogen receptor; IFN γ , interferon- γ ; M2, M2 anti-inflammatory macrophage; ROS, reactive oxygen species; TCA cycle, tricarboxylic acid cycle; T_H1 cell, T helper 1 cell; TNF, tumour necrosis factor.

along with the effects of sex chromosomes⁶². Trans individuals have unique chromosomal and endocrine profiles, and although some use cyclical hormone regimens, GAHT typically maintains long-term constant sex hormone levels that contrast to the cyclical hormonal changes observed in individuals assigned female at birth and the age-related declines seen in cisgender individuals⁶³. Therefore, GAHT could be expected to upregulate hormone receptor expression throughout the body, causing levels to surpass normal physiological levels on the basis of sex chromosomes and reproductive age. Hypothetically, this differential expression could modify immune and CNS functions, leading to changes in MS outcomes compared with cisgender individuals; however, this hypothesis remains untested. Research into gender-diverse dimorphisms in MS could reveal how sex hormones, independent of chromosomes, modulate immune function and neuroprotection, potentially informing the use of exogenous hormones to improve outcomes.

Exogenous sex hormones in MS

The effect of exogenous hormones in the context of MS, particularly in immune modulation and neuroprotection, is an emerging field of interest. Existing evidence is complex and predominantly informed by preclinical research. Furthermore, clinical studies are marked by diverse hormone therapy indications and doses, the inability to control for the aromatization of testosterone to oestradiol, and limitations such as small sample sizes and the absence of standardized measurements for MS progression and neurocognitive outcomes¹³. In the next sections, we discuss noteworthy trends observed in the available data that underscore the necessity for additional research, particularly through large, controlled trials.

Oestrogen

Oral contraceptive oestrogen. Low-dose cyclic oestrogen has long been thought to impact MS inflammation owing to the higher incidence of inflammatory diseases in cisgender women during their reproductive years⁶². One approach taken by several groups to understand the impact of exogenous hormones on MS outcomes has been to study the impact of combined oral contraceptives on inflammatory activity. One single-centre analysis ($n = 162$) of people assigned female at birth with relapse-onset MS using injectable disease-modifying therapies (DMTs) has found no difference in annualized relapse rate among individuals who had prior or current use of, or had never used oral contraceptives⁶⁴. The most commonly used contraceptive agents in the study were combinations of oestrogen and progestin, a synthetic form of progesterone⁶⁴. Another study of 495 people assigned female at birth with clinically isolated syndrome has found no association between use of oestrogen, progesterone or combination contraceptives and the risk of a second attack or disability accrual⁶⁵.

Although the studies above suggest little overall effect of combined oral contraceptives on inflammatory activity, a small retrospective study ($n = 46$) on continuous versus cyclic contraceptives (progestin and ethinyl oestradiol) has shown differences in MRI inflammatory activity, with the continuous group demonstrating a longer time to T2 lesion and contrast-enhancing lesion formation⁶⁶. Another multicentre trial, comprising 150 people assigned female at birth with relapse-onset MS treated with injectable DMTs, has found high-dose oral oestrogen contraceptive (40 µg ethinylestradiol and 125 µg desogestrel) use led to reduced active MRI lesions and better cognitive outcomes over 96 weeks compared with low-dose oestrogen (20 µg ethinylestradiol and 150 µg desogestrel) or no contraception⁶⁷.

Differences between the low dose and no oral contraceptive were negligible⁶⁷. These studies indicate that inflammatory activity is not increased with oral contraception and suggest that the sex hormones have anti-inflammatory effects in the CNS that potentially improve disease outcomes. However, the small sample sizes highlight the need for further research to confirm any true effect.

High-dose oestrogen. High (gestational) doses of oestrogens are widely recognized to for their anti-inflammatory effects and have been shown to improve experimental autoimmune encephalomyelitis (EAE) outcomes in both male and female animals. In mice, these high doses prevent EAE onset and modulate disease progression by inducing anti-inflammatory cytokine changes, shifting from T_H1 to T_H2 cells, increasing regulatory T and B cells, enhancing M2 macrophages and reducing demyelination^{59,68}.

One small study administering gestational oestrogen (oestriol) at 8 mg/day, mirroring concentrations seen at 6 months of gestation, to six individuals assigned female at birth with relapse-onset MS and four with secondary progressive MS has shown a reduction in the T_H1-produced cytokine IFN-γ, which is responsible for promoting inflammation^{18,69}. The study has further demonstrated a 79% reduction in total active gadolinium-enhancing MRI lesion volume and 82% reduction in lesion number during the 3 months of treatment, which returned to pre-treatment levels after oestrogen cessation in the relapse-onset MS group⁶⁹. During the 4-month retreatment extension phase, enhancing lesion volumes decreased again by 88%, and numbers decreased by 48% compared with the original baseline⁶⁹. The relationship did not reach statistical significance in participants with secondary progressive MS, in which inflammatory activity is lower and degenerative processes predominate⁶⁹.

In a randomized, placebo-controlled phase II trial, 164 people assigned female at birth with relapse-onset MS, who received 8 mg/day oestriol supplementation for 24 months alongside the DMT glatiramer acetate, demonstrated no reduction in annualized relapse rate or T2 lesion volume compared with glatiramer acetate alone⁷⁰. However, extended analysis have shown reduced cerebral cortex atrophy by month 12 in the oestriol group in individuals without baseline gad-enhancing lesions⁷¹. In addition, at month 12, the oestriol-treated group exhibited lower serum neurofilament levels, increased IL-5 and IL-10 levels, and decreased tumour necrosis factor (TNF), indicating a shift to a less inflammatory immune response^{71,72}. Subsequently, a subset of individuals from the same trial participated in a secondary 12-month phase II trial ($n = 62$ oestriol and $n = 49$ placebo) that showed improvements with oestriol in cognitive function, particularly processing speed and working memory assessed by the paced auditory serial addition test (PASAT), with spared frontal grey matter atrophy observed on MRI⁷³. These findings, coupled with the previous studies, suggest that oestriol offers potential neuroprotection in MS. However, further research is required to enhance the evidence base for oestrogen benefit.

Exogenous oestrogen is associated with adverse effects, including thromboembolism, endometrial hyperplasia and breast cancer^{74,75}. The risks vary depending on oestrogen type and the route of administration, and whether it is combined with progesterone^{74,75}. Although topical oestrogens generally have a safer profile than oral forms, oral oestrogens are often required to achieve gestational levels. Oestriol, used in the above-mentioned study⁷⁰, has a lower affinity for ERα in breast and uterine tissues, contributing to a better safety profile and lower cancer risk^{70,76}; irregular menses was the only side effect reported in the oestriol trial⁷⁰.

Menopause and menopausal hormone therapy. During menopause, declining oestrogen levels lead to the degradation of oestrogen receptors, which results in the suppression of neuroprotective pathways⁷⁷. Menopause in the general population is associated with increased cerebrovascular disease, and early menopause correlates with a fivefold higher risk of mortality from neurological disorders⁷⁸.

The effect of menopausal hormone therapy (MHT) on neurological function and cognition remains inconclusive. Some observational studies suggest protective effects on cognition and dementia-associated neuropathology^{79–81}. However, the Women's Health Initiative Memory Study (WHIMS) reported an overall increase in dementia risk after MHT in women over 65 years⁸². Further analysis has found no cognitive effects in people aged 50–55 years⁸³. Similarly, larger controlled studies have found neither harm nor benefit in cognitive function regardless of MHT timing and a favourable effect on menopause symptoms^{84,85}. Moreover, MHT has been shown to protect against global ischaemia-induced neuronal cell death and reduced carotid artery intima-media thickness in early menopause but not late menopause⁸⁶. These findings suggest a neuroprotective window for MHT that might offer secondary beneficial effects on MS outcomes, warranting further investigation.

Limited retrospective observational studies suggest a possible exacerbation of MS symptoms following menopause^{13,87,88}. One study has suggested better physical quality of life in people with MS using MHT⁸⁷. However, a study of people assigned female at birth from the Danish MS registry ($n = 3,325$), 10% of whom were using MHT, has found no association between disability accrual and MHT use⁸⁹. Two small interventional studies have assessed the effect of MHT in people with MS. The first study has randomly assigned 24 cisgender women with MS post-menopause to receive oestrogen and bazedoxifene, a selective oestrogen receptor modulator, or placebo⁹⁰. The second open-label study has compared 1 or 2 mg of oestradiol and cyclical 10 mg dydrogesterone treatment for 1 year in 14 individuals assigned female with relapse-onset MS stable on moderate efficacy DMTs with 13 healthy controls⁹¹. Both studies have shown a reduction in vasomotor symptoms in the MS groups without any safety concerns or worsening of disease and the second study showed improvement in mood and insomnia^{90,91}. No studies to date have assessed the use of MHT and inflammatory MS outcomes. Moreover, the distinction between effects of menopause and those from ageing, both of which can affect MS, raises a considerable challenge. Future research could benefit from including age-matched individuals assigned male to more accurately differentiate between ageing and menopause.

Selective oestrogen receptor modulators. Selective oestrogen receptor modulators (SERMS) can act as agonists or antagonists to oestrogen receptors depending on tissue type and are used to treat certain breast cancers and to prevent and treat osteoporosis in postmenopausal women, and they can be used as an alternative to traditional MHT. Some evidence suggests that SERMS can induce neuroprotective effects⁹², and mixed effects on cognition have been reported in non-MS populations; however, the results are complicated by large heterogeneity in cohorts and comparisons between people with and without breast cancer^{92–95}. Studies that explore the link between tamoxifen and dementia risk also show mixed results^{96–98}. In preclinical studies, certain SERMs, including lasofoxifene, bazedoxifene and tamoxifen, have shown promise for myelin repair and oligodendrocyte progenitor cell differentiation, which is crucial for remyelination potential^{99,100}. The Re-WRAP double-blind randomized controlled

clinical trial (NCT04002934) investigating the remyelinating effects of bazedoxifene in postmenopausal women with MS is underway¹⁰¹.

Progesterone

Progesterone promotes immune protection during pregnancy by modulating immune cell activity, which exerts anti-inflammatory effects that induce immune tolerance and reduce axonal damage¹⁰² (Fig. 1). Progesterone has been shown in numerous animal studies to have both myelinating and pro-myelinating effects that are thought to be mediated by progesterone receptor signalling and its impact on inflammation¹⁰³.

An observational study ($n = 30$) has found that low progesterone relative to oestrogens was linked to an increase in gadolinium enhancing lesions on MRI in people assigned female at birth¹⁰⁴. The POPART-MUS placebo-controlled trial, investigating the anti-inflammatory potential of a high-dose progestin (10 mg/day 19-nor-progesterone derivative norgestrel acetate) combined with transdermal 17- β -oestradiol (75 μ g, once a week) in individuals with relapse-onset or secondary progressive MS, did not show a significant effect on the annualized relapse rate at the 12-week follow-up. This study was terminated early owing to poor recruitment, and the 12-week trial duration tend to be insufficient to accurately measure annualized relapse rates, particularly as relapses are becoming less common in the era of effective DMTs for MS¹⁰⁵.

Prolactin

Prolactin, primarily secreted by the anterior pituitary to promote lactation, is also produced by extra-pituitary tissues and is involved in immune modulation, B cell proliferation and neuroregeneration¹⁴. Prolactin potentially has a dual role in MS, contributing to immune system activation alongside neuroprotection and remyelination^{106,107}. Several studies and meta-analyses have reported elevated prolactin levels in people with MS, with some associating these elevations with increased relapse rates and others with reduced disability, highlighting the current inconclusiveness of the evidence¹⁰⁸. Hyperprolactinaemia is not specific to MS, with elevated levels observed in other autoimmune conditions, psychosocial stress and dysregulation of the hypothalamic–pituitary–adrenal axis¹⁴.

Prolactin exhibits dose-dependent effects in EAE models, with low doses showing no detrimental effects on the disease and high doses showing exacerbated severity¹⁰⁹. One open label trial involving 18 individuals with MS has used bromocriptine (2.5 mg twice daily) to reduce prolactin levels. The treatment did not halt disease progression, nor was exacerbation of MS reported, indicating no effect on MS outcomes¹¹⁰.

Assessment of MS outcomes during breastfeeding, when prolactin levels are high, is complex. Although results from a meta-analysis suggest that breastfeeding might reduce postpartum relapse risk, this finding could be confounded by study biases¹¹¹. People with highly active MS who are at greater risk of relapse have higher tendency to resume DMTs and stop breastfeeding postpartum, whereas individuals with lower disease activity might breastfeed longer and experience fewer relapses¹¹². Furthermore, a reduced relapse rate in breastfeeding might also be attributed to breastfeeding-associated amenorrhoea and the modulation of endogenous oestrogen.

Androgens

Both testosterone and DHT exhibit neuroprotective and trophic effects on neurons and safeguard against glutamate-induced toxicity and oxidative stress^{113,114}. Studies in vitro and in EAE models indicate

that testosterone treatment can promote a T_H1 -to- T_H2 shift, leading to reduced pro-inflammatory cytokine production and increased neuroprotective and pro-myelination properties^{75,115,116}.

Androgens in people assigned male at birth. Hypogonadism is a known risk factor for autoimmunity and could have implications in MS. In the largest longitudinal observational study involving 96 people assigned male at birth (mean age of 40 years) with MS or clinically isolated syndrome, 39% had central hypogonadism¹¹⁷. The study also found that lower age-adjusted endogenous testosterone (<288 ng/dl) correlated with more severe cognitive scores on the symbol digit modalities test and worse disability outcomes measured by Expanded Disability Status Scale score¹¹⁷. A 2024 systematic review looked at the effects of hypogonadism in people assigned male at birth who have MS: 11 out of 19 studies reported a positive connection between low testosterone and hypothalamic–pituitary–gonadal axis dysfunction¹¹⁸. The review also found a positive association between low testosterone and sexual dysfunction¹¹⁸. The data indicate an association between low testosterone and MS, with further research needed to determine causality and the effects of testosterone replacement on outcomes.

Studies that investigate the use of exogenous testosterone as a potential MS therapy are limited¹¹⁹. A small pilot study of cisgender men with relapse-onset MS not on DMTs ($n = 10$, mean age of 46 years) has tested the effects of testosterone treatment on disease pathology and cognition¹²⁰. The results showed an improvement in cognition measured by PASAT and a 67% reduction in brain volume loss over 9 months of treatment compared with the pre-treatment period. No effect was found on inflammatory relapses or MRI activity¹²⁰. Subsequent analysis of this trial has also found a significant increase in grey matter within the right frontal cortex with testosterone therapy compared with pre-treatment MRI¹²¹. Furthermore, participants were found to have reduced immune mediated inflammation indicated by the delayed-type hypersensitivity skin recall test, reduced inflammatory cytokine production and increased brain-derived neurotrophic factor (a potential marker of neuroprotection)¹²². Collectively, these findings suggest that testosterone exhibits dual anti-inflammatory and neuroprotective properties that could benefit MS. The TOTEM-RRMS randomized, controlled trial investigating testosterone replacement in cisgender men with hypogonadism and relapse-onset MS currently treated with natalizumab is in progress (NCT03910738)¹²³.

Androgens in people assigned female at birth. An observational study investigating serum sex hormones in people assigned female at birth with relapse-onset MS has found lower testosterone levels compared with healthy controls⁵⁸. Furthermore, low testosterone correlated with heightened inflammatory MRI changes and disability⁵⁸. A larger cross-sectional study has found no significant relationship between testosterone levels and MS outcomes for either sex¹²⁴. These contradictory results could reflect differences in study populations, reproductive age, MS course, and variability in hormone levels and measurement methods, underscoring the need for large controlled trials.

Fertility treatments

Historically, five small trials have suggested an increased risk of MS relapses with assisted reproduction, particularly with gonadotrophin-releasing hormone analogues (GnRHa), which cause a sudden decline in sex hormones¹²⁵. More recent studies combining DMTs with fertility treatments, including GnRHa, do not replicate the

relapse risk seen in earlier trials lacking DMTs, suggesting that judicious use of DMTs can help to prevent relapses^{126,127}.

GAHT in MS

Our current understanding of GAHT and the immune system

GAHT regimens depend on the goals and desired physical changes of an individual, alongside local guidelines, therapeutic availability and accessibility (Fig. 3). Gender affirmation can encompass gender-affirming surgery, which can involve removal of gonadal organs (oophorectomy or orchidectomy). In feminizing hormone therapy, trans women and non-binary people assigned male at birth are typically prescribed oestradiol and anti-androgens to suppress testosterone levels and promote the development of feminine characteristics¹²⁸. In masculinizing hormone therapy, trans men and non-binary people assigned female at birth are typically prescribed testosterone of standard or low doses to induce masculine changes. Younger individuals might be prescribed GnRHa in the early stages of puberty to prevent undesired sex characteristics developing prior to commencing GAHT. Guidelines for GAHT recommend regular blood analysis to ensure that hormone levels are maintained within reference ranges of the affirmed gender¹²⁹.

In contrast to cisgender individuals, trans people using GAHT exhibit a unique chromosome–endocrine profile, which might help us to better understand how sex hormones influence autoimmunity. The X chromosome is abundant in genes that influence immune regulation, such as *TLR*, *CD40*, *CXCR3* and *FOXP3* (refs. 130–132). A 2022 study on the effect of GAHT on DNA methylation in trans people has demonstrated that methylation patterns, including those coding for immune promotor genes, often mirror those of cisgender individuals of the same affirmed gender¹³³. This finding suggests that although some immune functions are chromosome-dependent and unaffected by GAHT, GAHT can alter DNA methylation, which in turn can influence other non-permanent immune functions. Limited research to date confirms this theory, with some immune functions changing whereas others remain unchanged^{41,134–141} (Box 2).

Elevation in inflammatory markers has been observed in both trans women receiving exogenous oestrogen and trans men using exogenous testosterone^{135,142}. Oestradiol and anti-androgens increase chemokine expression, which affects lymphocyte migration patterns and suppresses natural killer cell activity^{137,143}. Sex-based differences in immunoglobulin and antibodies mostly persist after GAHT, whereas regulatory T cell expression increases in trans men using GAHT in line with sex differences known to exist between cisgender men and women^{143–145}. In trans women, oestradiol can deviate T cell differentiation towards T_H2 cells, whereas the opposite effect can be observed with testosterone in trans men^{41,138,139}. At present, the long-term significance of these immune alterations remains uncertain owing to a lack of studies with comprehensive follow-up, although recognition of the still-nascent nature of this field of study is important.

GAHT in people with MS

Very little is known about the impact of GAHT on MS outcomes. Research in this field is limited by difficulties in recruiting adequate sample sizes for a rare disease within a small population. The heterogeneity in MS and the complexities of GAHT, encompassing different therapeutic regimes alongside a diversity in hormone levels across individuals and ages, present considerable challenges. Furthermore, understanding of these complexities often exceeds the current scope of knowledge among neurologists. The absence of gender identity and

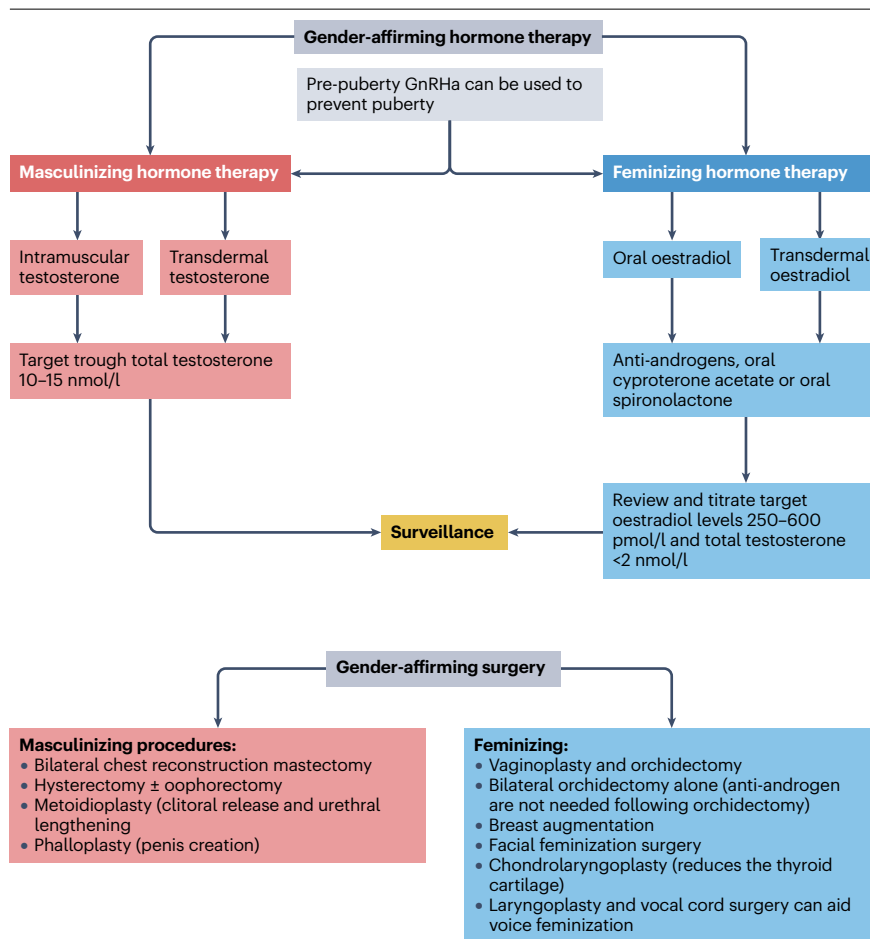


Fig. 3 | Gender-affirming therapy. An overview of gender-affirming therapy. Transgender individuals can affirm their gender in many ways, including changing names, pronouns or gender expression and using gender-affirming hormone therapy and/or surgery. Gender-affirming therapy varies by patient preference, local policies and access to therapies. GnRHa, gonadotrophin-releasing hormone analogue. Figure adapted with permission from ref. 63, Wiley/© 2019 AMPCo Pty Ltd.

hormone level data in existing MS registries further limits the ability to comprehensively understand the impact of GAHT on MS outcomes.

One retrospective national record-linkage study in Britain that compared individuals with gender identity disorders and cisgender controls has found an elevated adjusted relative risk of MS in trans women of 6.63 (95% CI 1.81–17.01, $P = 0.0002$)¹⁴⁶. This risk was based on four observed cases compared with the expected 0.6 cases. No increased risk of MS was found in trans men compared with cisgender women, with an adjusted relative risk of 1.44 (95% CI 0.47–3.37, $P = 0.58$), based on five observed cases in trans men and 3.5 expected cases¹⁴⁶. The study had substantial limitations, including insufficient details on GAT and the temporal relationship with MS diagnosis. The study also included adolescents who can affirm their gender without GAHT or use puberty blockers instead of GAHT. Therefore, the conclusion that androgen-blocking therapies and exogenous oestrogen use in trans women might be associated with increased risk of MS should be interpreted with caution.

The long-term impact of GAHT on brain structure, function and cognition remains uncertain, again owing to a lack of comprehensive long-term studies. Oestrogen is linked with improved verbal memory in trans women without MS^{134,147}. Preliminary small-scale MRI studies have shown an increase in cortical thickness and alterations in the neural structures of the hypothalamus following exogenous hormones in trans people, suggesting that treatment induces structural changes

towards proportions that resemble those of their affirmed gender^{148,149}. These findings suggest that exogenous hormones can affect brain structure, including cognitive areas, but the clinical significance is unclear and requires further research.

Clinical considerations for people with MS using GAHT

Sexual and gender minority (LGBT+) individuals often face difficulties in health care owing to systemic disparities, discrimination, including from health-care providers, and a shortage of culturally competent care^{150–154}. The 2015 US Transgender Survey, the largest of its kind comprising 27,715 trans people, found that 33% of individuals who saw a clinician in the past year faced negative experiences related to their gender identity¹⁵⁵. Additionally, 23% avoided medical care because of fear of mistreatment, which could lead to more advanced illnesses and a greater need for urgent care¹⁵⁵.

A Northern California study of 26 LGBT+ participants (15% trans, non-binary or other) has found that most respondents felt that their sexual orientation or gender identity (SOGI) did not affect their MS care, although the location of the study in a region known for LGBT+ activism might limit its broader applicability¹⁵⁴. Another North American MS-specific observational study of 5,604 participants has found that SOGI was associated with lower health-care satisfaction and less comfort discussing sexual health, but no significant differences in health care use were observed³. However, the generalizability of these

findings tend to be limited as only 25 transgender individuals were included in the study.

Health-care disparities for trans people result from avoidance of care, alongside insufficient provider knowledge of trans health owing to inadequate formal education on the subject¹⁰. A study of American Academy of Neurology members found that although most neurologists recognized general barriers faced by LGBT+ individuals and felt comfortable assessing people from this population, many did not understand the interplay between SOGI and GAHT on neurological conditions¹⁵⁶. Guidelines for trans health care emphasize that providers must acquire the necessary competencies to deliver patient-centred, affirming care, and these competencies should be incorporated into neurology practice^{10,157}.

Management of the clinical care of trans people with MS involves key clinical considerations outlined in Box 3. Beyond the use of MS DMTs, evidence demonstrates that lifestyle factors impact MS outcomes¹⁵⁸. As such, addressing comorbidities and adopting brain-healthy lifestyles, such as smoking cessation, maintaining healthy

weight and increased physical exercise, improve MS outcomes^{158,159}. Evidence suggests GAHT can increase the risk of various comorbidities, including stroke, malignancy, osteoporosis, cardiovascular issues, migraine and fatigue^{157,160,161}. Trans people also experience higher rates of mental and physical illnesses, such as depression, chronic pain, functional neurological disability, obesity and psychosocial stressors^{154,162–164}. Health experiences and clinical trajectories of trans people can differ considerably from cisgender individuals, often resulting in higher disability rates^{157,165}. Additionally, the incidence of neurodiversity is notably higher in the trans population, highlighting the need for tailored health care that addresses the unique needs of this patient group^{166–168}.

When combining DMTs with GAHT, thorough safety surveillance is essential. The authors believe that any potential risk of inflammatory MS activity associated with GAHT tends to be mitigated with judicious use of DMTs, similar to findings from the aforementioned fertility studies (see the section ‘Fertility treatments’). We also believe that GAHT does not tend to considerably affect DMT efficacy. In the absence of

Box 2 | Summary of GAHT effects on the immune system

Elevation of inflammatory markers

Transgender (trans) men

- 47 trans men using transdermal testosterone showed elevated C-reactive protein (CRP) levels 12 months after gender-affirming hormone therapy (GAHT) initiation¹⁴²
- 23 trans men using testosterone attenuated type I interferon activity and enhanced tumour necrosis factor (TNF), interleukin and interferon- γ responses²³⁰
- Testosterone increased inflammatory markers TNF and CXC-chemokine ligand 1 (CXCL1) but not CC-chemokine ligand 2 (CCL2)²³¹

Trans women

- Oral oestrogen increased CRP, IL-1, IL-6, IL-8 and TNF in trans women^{134,135}
- 49 trans women using transdermal oestradiol with cyproterone acetate did not show increased CRP levels 12 months after GAHT initiation¹⁴²

Influence on lymphocyte behaviour

Trans men

- Testosterone prompts a shift in peripheral blood T cells towards a T helper 1 (T_H1) pathway, whereas oestrogen skews the differentiation towards a T helper 2 (T_H2) response¹³⁷

Trans women

- Oestrogen in combination with anti-androgens significantly increases the expression of CC-chemokine receptor 1 (CCR1), CXC-chemokine receptor 3 (CXCR3) and CCR5, affecting lymphocyte migration patterns, and suppresses the numbers and cytotoxic activity of natural killer cells^{137,150}

Differences in immune globulin levels

Trans men

- Cisgender women generally have higher levels of circulating IgG and IgM compared with cisgender men; one study has

found that baseline serum IgG and IgM levels were significantly higher in trans men than in trans women; GAHT induced only minor changes in antibody levels after 4 and 12 months, with no reversal of initial gender-related differences in antibody levels^{143–145}.

Trans women

- Baseline serum IgG and IgM levels were significantly higher in trans men than in trans women¹⁴³
- GAHT induced only minor changes in antibody levels after 4 and 12 months, with no reversal of initial gender-related differences in antibody levels^{143–145}

T_{reg} cell expression

Trans men

- Regulatory T (T_{reg}) cell numbers are higher in cisgender and trans men than in cisgender and trans women; GAHT changed T_{reg} cell gene expression in trans men including upregulation of cytokine-mediated signalling and secretion pathways^{41,138,139}
- GAHT increases the expression of genes related to T_{reg} cell function in trans men compared with cisgender and trans women but not to the suppressive levels observed in cisgender men⁴¹

Trans women

- GAHT caused changes in T_{reg} cell gene expression in trans women with downregulation in cell growth and activation pathways^{41,138,139}

ANA

Trans men

- 36% of trans men were anti-nuclear antibodies (ANA) positive compared with 13% in cisgender people^{140,141}

Trans women

- 31% of trans women were ANA positive compared with 13% in cisgender people^{140,141}

evidence for this combination, patients should undergo appropriate clinical and radiological surveillance, including routine monitoring of full blood counts, lipid profiles, glucose levels and clotting factors, which are well-established components of comprehensive MS care models¹⁰.

The risk of malignancy for people using DMTs or GAHT is thought to be low; however, the risk with combined use is uncertain^{169,170}. Therefore, comprehensive and individualized cancer screening should be considered in all patients on dual treatment. The influence of GAHT on cardiovascular health also warrants additional attention in clinical settings; good vascular health improves MS outcomes, particularly brain volume and cognition^{171,172}. Exogenous sex hormones oestradiol and testosterone each have both positive and negative effects on vascular risk factors^{157,173–176}. Cisgender women benefit from exogenous oestrogen and MHT to prevent osteoporosis, whereas GAHT paradoxically increases osteoporosis risk in trans women owing to hormone imbalances¹⁷⁷. Thus, monitoring and addressing bone health in trans people with MS, especially those at greater risk of falls, is important. More research is needed to fully understand any interactions with GAHT, ageing and comorbidities in people with MS¹⁷⁸.

Conclusion — the imperative of inclusive MS research

Here, we highlighted the considerable knowledge gap regarding the impact of GAHT on MS outcomes and clinical care, with only a few small studies to date addressing this issue. A pressing need exists to understand how GAHT influences MS outcomes and how best to manage the clinical care of trans people with MS. The impact of GAHT on disease outcomes for trans people with MS, including inflammatory, progressive and cognitive aspects, is currently largely unknown.

We have used the broader sex hormone literature to inform the role of endogenous and exogenous hormones in the immune system and CNS. Although knowledge gaps remain, a well-established interplay is evident between sex hormones and MS, highlighting their potential impact on both the inflammatory and neurodegenerative components of the disease. The use of exogenous sex hormones to modify the course of MS has a long history, although most trials are dated and under-powered; thus, robust data are lacking.

The International Advisory Committee on Clinical Trials in MS published guidelines in 2023 that advocated for increased diversity, including gender diversity, in research, to more accurately represent real-world populations and uncover demographic and non-demographic differences that could impact clinical care^{179,180}. Inclusion of trans people in MS research is crucial to strengthen this field and can provide important insights given their unique chromosomal and endocrine profiles that deviate from cisgender individuals. Currently, trans people are largely invisible in MS research. Demographic data collection must include two questions to ask, firstly, about gender and, secondly, about sex assigned at birth. Only with this two-step approach will both cisgender and trans people be accurately recorded and included^{179,180}.

Exogenous hormones are usually administered long term to affirm gender, offering insights into the extended and non-age-adjusted outcomes of sex hormone exposure in MS. Insufficient data are available to determine whether androgen-blocking therapies and/or exogenous oestrogen increase the risk of MS in trans people. The impact of testosterone is also unclear, although limited evidence suggests potential neuroprotective effects. The exploration of the impact of GAHT on disease progression and potential neuroprotection remains an uncharted territory, with evolving outcomes anticipated to be of particular interest to the MS field.

Box 3 | GAHT and MS in the clinical setting

DMT efficacy

No evidence currently exists regarding any change in efficacy of disease-modifying therapies (DMTs) combined with gender-affirming hormone therapy (GAHT); with high efficacy therapy, GAHT can be reasonably assumed to have no significant effects.

DMT safety surveillance

Many laboratory parameters have sex-specific reference ranges. GAHT can lead to alteration in laboratory parameters encompassing full blood counts, lipid profiles, renal function and clotting factors, which are routinely performed as part of comprehensive health care for multiple sclerosis (MS)^{10,232,233}. In general, once transgender (trans) people are on established GAHT, the reference range of the affirmed gender should be used for interpretation²³².

Malignancy risk

Malignancy risk when combining DMTs and GAHT is unknown. Autoimmunity and DMT-related immunosuppression might potentially increase risk of malignancy in people with MS; however, evidence suggests this risk is low or absent, but further analysis is needed^{234–236}. The relationship between GAHT and risk of malignancy is a complex and evolving topic as risks have historically been

overestimated^{170,237}. The authors highlight the need for individualized cancer surveillance.

Vascular healthy

Cardiovascular disease can impact MS outcomes by contributing to systemic inflammation alongside impacting brain volume and cognitive outcomes^{171,172}. GAHT can have both positive and potentially negative effects on cardiovascular risk factors. Exogenous oestrogen might have a favourable effect on lipid profiles; however, oestrogen is associated with an increased risk of thromboembolism including stroke, with an established increased stroke risk and myocardial infarction risk in trans women^{157,173}. Testosterone can increase muscle mass and reduce body fat, which can then improve insulin sensitivity and lipid profiles. But other evidence shows that testosterone can have an adverse effect on lipid profiles and contribute to polycythaemia^{174–176}.

Osteoporosis

Some evidence suggests compromised bone microarchitecture in trans women, particularly if oestradiol concentrations are low or insufficient^{238,239}. Bone health does not appear to be compromised in trans men; testosterone can improve bone mineral density²³⁹.

Moving forward, the first and most important step is for existing databases to integrate and collect data on gender identity and sex assigned at birth to recognize trans individuals. Collaborative international studies, both retrospective and ideally prospective, should compare MS outcomes in trans people on GAHT compared with those who are not on GAHT, or with matched cisgender controls, although we acknowledge that the heterogeneity of endocrine profiles in both groups can complicate matching. In the rapidly evolving field of medicine, inclusive practice is not merely a reflection of current trends but an obligation to provide the highest standard of care for all patients.

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Author contributions

C.N. researched data for the article. All authors contributed substantially to discussion of the content. C.N., A.S.C. and V.G.J. wrote the article. All authors reviewed and/or edited the manuscript before submission.

Competing interests

A.S.C. has received product from Besins Healthcare for investigator-initiated clinical studies using oestradiol and progesterone. No monetary support from Besins Healthcare has been received for any studies, and Besins Healthcare have had no input into study design, data analysis or writing of any manuscripts.

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